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Phylogenetic tree statistics: A systematic overview using the new R package 'treestats'

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ABSTRACT

Phylogenetic trees are believed to contain a wealth of information on diversification processes. However, comparing phylogenetic trees is not straightforward due to their high dimensionality. Researchers have therefore defined a wide range of low-dimensional summary statistics. Currently, it remains unexplored to what extent these summary statistics cover the same underlying information and what summary statistics best explain observed variation across phylogenies. Furthermore, a large subset of available summary statistics focusses on measuring the topological features of a phylogenetic tree, but are often only explored at the extreme edge cases of the fully balanced or imbalanced tree and not for trees of intermediate balance.

Here, we introduce a new R package called 'treestats', that provides speed optimized code to compute 70 summary statistics. We study correlations between summary statistics on empirical trees and on trees simulated using several diversification models. Furthermore, we introduce an algorithm to create intermediately balanced trees in a well-defined manner, in order to explore variation in summary statistics across a balance gradient.

We find that almost all summary statistics are correlated with tree size, and find that it is difficult, if not impossible, to correct for tree size, unless the tree generating model is known. Furthermore, we find that across empirical and simulated trees, at least three large clusters of correlated summary statistics can be found, where statistics group together based on information used (topology or branching times). However, the finer grained correlation structure appears to depend strongly on either the taxonomic group studied (in empirical studies) or the tree generating model (in simulation studies).

Amongst statistics describing the (im)balance of a tree, we find that almost all statistics vary non-linearly, and sometimes even non-monotonically, with our generated balance gradient. This indicates that balance is perhaps a more complex property of a tree than previously thought. Furthermore, using our new imbalancing algorithm, we devise a numerical test to identify balance statistics, and identify several statistics as balance statistics that were not previously considered as such. Lastly, our results lead to several recommendations on which statistics to select when analyzing and comparing phylogenetic trees.

1. Introduction

Phylogenetic trees contain information on diversification in both the way nodes and tips are connected to each other (e.g. their topology) and in the sequence of branching events (e.g. the branching times). This source of information has been used extensively, for instance to infer underlying evolutionary processes such as speciation (Nee, 2001), extinction (Nee & Holmes, 1994), the interaction between traits and diversification (Herrera-Alsina et al., 2018), to reconstruct ancestral states (Pagel et al., 2004; Yu et al., 2015) and within the field of phylogenetic comparative methods (Revell & Harmon, 2022).

However, phylogenetic trees are hard to compare directly with each

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other due to their high dimensionality. Comparison between trees is therefore often made using lower dimensional summary statistics focusing on a subset of all available summary statistics, in order to capture as much information as possible. This ranges from looking only at a single statistic (often a branch-times related statistic) (Guimarães Fabreti & Höhna, 2023; Harmon et al., 2003; Liow et al., 2010), combining a topology-based and branch-times based statistic in an effort to cover as much information as possible with a very small selection of statistics (Hagen et al., 2015; Janzen & Etienne, 2017; Verboom et al., 2020), to using a large number of summary statistics in Approximate Bayesian Computation (Saulnier et al., 2017), Machine Learning (Ruffley et al., 2019) and Deep Learning (Lambert et al., 2023; Voznica et al., 2022; Qin et al., 2024). However, it remains unclear to what extent choices of used summary statistics are impacted by potential correlations and information overlap between statistics. An important first step towards an increased understanding has been taken by Tucker and colleagues, who investigated the choice of summary statistic for community ecology related studies, and identified several 'anchor' summary statistics, representing three important dimensions of a phylogenetic tree: richness, divergence and regularity (Tucker et al., 2017). For richness, they selected phylogenetic diversity; for divergence the selected mean pairwise distance and for regularity they chose variance of pairwise distance. However, their approach was geared towards community assembly, also taking into account species abundances at the tips (and often ignoring zero-abundance species), whereas we focus on clade diversification.

A subset of all summary statistics focuses on measuring the balance of a tree (see for a comprehensive review (Fischer et al., 2023)). Balance of a tree reflects the distribution of daughter tips in different parts of the phylogenetic tree, mimicking the physical act of balancing a tree. Generally, there is consensus on what constitutes a fully balanced tree: where each branch bifurcates into two daughter branches until reaching the present time. Analogously, a fully imbalanced tree reflects a tree where all daughter branches stem from the same parental lineage, such that all branches do not have their own daughter branches, generating a so-called 'caterpillar', 'ladder'-like or 'comb' tree (Fig. 1). Balance statistics then, maximize their value on a fully balanced tree, and minimize their value on a fully imbalanced tree. Similarly, imbalance statistics do the inverse. What remains unexplored however, is how summary statistics capture intermediately balanced trees, e.g. can we use summary statistics to identify how close we are to a fully (im)balanced tree? Furthermore, it remains unclear to what extent different balance metrics measure the same aspects of balance of a phylogenetic tree.

Here, we introduce the R package treestats, that computes 69 previously proposed summary statistics for phylogenetic trees, allowing for a quick and user-friendly analysis of the interrelationship between summary statistics. We explore and investigate correlations between these summary statistics on both empirical and simulated phylogenetic trees. Furthermore, the package computes a new balance statistic (the imbalance steps statistic) that we introduce here, alongside algorithms to generate trees of intermediate balance. We use these algorithms to study how balance statistics vary with varying degrees of intermediate balance.

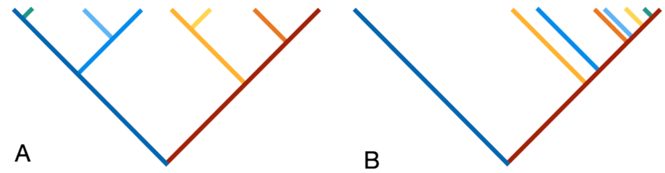

Fig. 1. A fully balanced (A) or fully imbalanced (B) tree of 8 tips, highlighting the two crown lineages (blue and red shades), and their respective daughter lineages, where daughter lineages are colored depending on their parent lineage (e.g. all daughter lineages stemming from the blue crown lineage are colored in a blue-spectrum color). Both trees have the same sequence of branching times and same branch lengths, and tree B has been constructed from tree A by re-attaching all branches to the dark red crown lineage. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2. Methods

We introduce the R package 'treestats', which computes 70 summary statistics for phylogenetic trees. We focus here on statistics applying to ultrametric, time-calibrated, strictly bifurcating phylogenetic trees. The vast majority of developed statistics applies to these trees, but we do want to point out to the reader that there are several statistics available that apply specifically to non-time calibrated trees, making use of differences in substitutions across branches in the tree (Mongiardino Koch, 2021; Smith et al., 2018; Steenwyk et al., 2021), and hence these statistics are not considered here.

For each statistic, we developed a fast (Figure S3), C++ based implementation from scratch, that generates a speed improvement of often several orders of magnitude (Figure S4). The R package was created using test-driven development, where the new implementations of summary statistics were rigourously tested against already existing code of such statistics in other R packages. Currently (treestats version 1.1.2), 98 % of all code is tested for correctness. The remaining 2 % includes code that ensures fail safes against incorrect data entry, which cannot be fully anticipated, and therefore cannot be fully tested.

2.1. Summary statistics

We describe each summary statistic, where we classify them using the general source or type of information used per statistic (see Table 1).

2.1.1. Node statistics

Firstly, we collected a number of statistics summarizing properties of nodes, in strictly binary, bifurcating, trees. Each node indicates the branching event into two daughter lineages. Here, we use the notation 'L' and 'R' to indicate the number of tips attached to the 'Left' and 'Right' daughter branch. Summing over all nodes in the tree, we have the following statistics, indicating the calculation per node:

1. Colless index, abs (L – R) (Colless, 1982).
2. Corrected Colless index, abs(L-R) / ((n – 1)(n-2)) (Heard, 1992).
3. Quadratic Colless index, $(L - R)^2$ (Bartoszek et al., 2021).
4. Blum statistic, log(L+R-1) (Blum & François, 2006).

5. Rogers J index, L!= R (Rogers, 1996).
6. equal weights Colless index, abs(L – R) / (L+R – 2) (Mooers & S. B. Heard, 1997).
7. Number of pitchforks, L+R=3 (Kendall et al., 2018).
8. Number of cherries, L+R=2 (McKenzie et al., 1999).
9. Number of double cherries, L=2 and R=2 (Chindelevitch et al., 2021).
10. Fourprong, the number of 4-tip caterpillars, L=3 and R=1 or R=3 and L=1 (Chindelevitch et al., 2021).
11. Stairs measure, the fraction of nodes with (L!= R) (Norström et al., 2012).
12. Stairs2 measure, the average of min(L, R) / max(L, R) (Norström et al., 2012),
13. ILnumber, nodes with a single tip child, e.g. either L=1 or R=1 (Kendall et al., 2018).
14. Rquartet index, $\binom{L}{2}+\binom{R}{2}$ (Coronado et al., 2019).
15. I statistic, (max(L, R) – (L+R) / 2) / (L+R – 1 – (L+R) / 2) (Fusco & Cronk, 1995; Purvis et al., 2002).
16. J^1, −(L/(L+R)) log(L/(L+R)) (Lemant et al., 2022).
17. Beta (beta_dist(L, R)) (Aldous, 1996).
18. Average ladder measure, fraction of nodes with either R=1 or L=1 and that are connected with each other (e.g. forming a ladder) (Kendall et al., 2018).
19. Maximum ladder statistic, longest chain of connected nodes with either R=1 or L=1 (e.g. forming the longest ladder) (Kendall et al., 2018).

Measuring only at the root, we have:

20. Root Imbalance: R / (L+R) with R>L (Guyer & Slowinski, 1993).

2.1.2. Depth statistics

The depth of a node (or tip) is defined as the distance between the root (or crown) and that node (or tip). Statistics using the depth are:

21. Maximum depth, measuring the maximum path (distance) between the tips and the root (Colijn & Gardy, 2014).
22. Sackin index, the sum of depths across all tips (Sackin, 1972).
23. Average leaf depth, the average depth of all tips (Shao & Sokal, 1990).
24. Total path length, the sum of depths across all tips and nodes (Colijn & Gardy, 2014).
25. Total internal path length, the sum of depths across all nodes (Knuth, 1997).
26. Variance of leaves' depth, the variance in tip depths (Coronado et al., 2020).
27. Average vertex depth, the average depth across all tips and nodes (Herrada et al., 2011).
28. B1, the sum of the inverse of all node depths (Shao & Sokal, 1990).
29. B2, the sum of the depth of each node divided by 2^{depth} (Shao & Sokal, 1990).
30. Symmetry Nodes index, the number of internal nodes that are not symmetry nodes, e.g. nodes from which the two subtrees have the same tree shape. Tree shape is measured as the distribution of node depths across all descendent nodes (Kersting & Fischer, 2021).

The width of a tree represents the number of occurrences of each depth in a tree. Summary statistics making use of this are:

31. Maximum width, the most common depth (Colijn & Gardy, 2014).
32. Maximum difference in widths, the maximum of the differences in occurrences between consecutive widths, where the widths are ordered by size (Colijn & Gardy, 2014).
33. Mw over Md, e.g. the maximum width divided by the maximum depth (Colijn & Gardy, 2014).

2.1.3. Distance matrix

The distance matrix represents all pairwise distances between tips (and potentially nodes), either expressed in the sum of branch lengths, or in the number of edges required to traverse the shortest path between the pair. Summary statistics based on this distance matrix are:

34. Total cophenetic index, the sum of the distances to the lowest common ancestor between two tips, expressed in number of edges (Mir et al., 2013).
35. Area per pair index, the sum of the number of edges on the path between all pairs of tips (Lima et al., 2020).
36. Mean pairwise distance (mpd), the mean distance between all pairs of tips in units of branch length (Tsirogiannis et al., 2012; Webb et al., 2002).
37. Variance in pairwise distance statistic, the variance in distance between all pairs of tips in units of branch length (Webb et al., 2002).
38. J statistic, the intensive quadratic entropy given by the average distance between two randomly chosen species, calculated as the sum of all pairwise distances divided by S^2, where S is the number of tips of the tree (Izsák & Papp, 2000).
39. Mean nearest taxon distance, the mean distance to the nearest tip (Webb et al., 2002).
40. Phylogenetic species variability index (psv), the sum of all pairwise relationships d_{ij}, where $d_{ij} = 0.5 * (d_{ii} + d_{jj} + d_{ij})$, where d_{xx} indicates the distance of tip x to the root (Helmus et al., 2007).

2.1.4. Eigen decompositions

The distance matrix also serves as a starting point for measuring the distance Laplacian spectrum. The starting point here is the distance Laplacian matrix, which is the distance matrix of the phylogenetic tree, where the diagonal zero entries are replaced by the negative row sums (e.g. the distance matrix is subtracted from a diagonal matrix formed by it's row sums).

Using the distance Laplacian matrix, the distance Laplacian spectrum statistic calculates the eigenvalues and eigenvectors of the distance Laplacian matrix (Lewitus & Morlon, 2016). Following the work by Lewitus & Morlon, we use four properties here:

41. Principal eigenvalue of the distance Laplacian matrix (laplace_spectrum_e in all figures).

42. Eigen gap of the distance Laplacian matrix, e.g., the position of the largest difference between eigen values, indicating the number of modalities in the tree (laplace_spectrum_g in all figures).
43. Asymmetry of the distribution of eigenvalues of the distance Laplacian matrix (laplace_spectrum_a in all figures).
44. Peakedness of the eigenvalues distribution of the distance Laplacian matrix (laplace_spectrum_p in all figures).

Instead of using the distance Laplacian matrix, two other statistics make use of the raw Laplacian matrix, e.g. before correcting the diagonal. The eigenvalues of the Laplacian matrix are thought to be related to the balance and conductance of a graph (Chindelevitch et al., 2021).

45. Minimum positive eigenvalue of the Laplacian matrix.
46. Maximum eigenvalue of the Laplacian matrix.

The eigenvalues of the adjacency matrix, e.g. the square matrix that indicates the connections between nodes and tips, contain information on the number of node-disjoint collections of edges in a phylogeny:

47. Minimum positive eigenvalue of the adjacency matrix.
48. Maximum eigenvalue of the adjacency matrix.

2.1.5. Network science

Summary statistics that take inspiration from network science, consider phylogenetic trees in their unrooted form, and use network properties to calculate properties of the trees. These are:

49. Wiener index, the sum of the shortest path lengths across all tips of the tree (Chindelevitch et al., 2021).
50. Diameter, the maximum of all shortest paths between all tips of the tree (Chindelevitch et al., 2021). When taking branch lengths into account, this is equal to twice the crown age in an extant tree (see below).
51. Maximum betweenness, the maximum value of betweenness for each node u, where betweenness is measured as the shortest path between nodes v and w, going through focal node u (Chindelevitch et al., 2021).
52. Maximum closeness, the inverse of farness, where farness is the sum of distances from a node to all the other nodes in the tree (Chindelevitch et al., 2021).
53. Maximum closenessW, same as maximum closeness, but distances are weighed by branch lengths (Chindelevitch et al., 2021).
54. Eigen centrality, the maximum value of the Perron-Frobenius eigenvector of the adjacency matrix of the tree (Chindelevitch et al., 2021).
55. Eigen centralityW, same as Eigen centrality, but distances are weighed by branch lengths (Chindelevitch et al., 2021).

2.1.6. Branching times

The sequence of branching times mainly holds information about the timing of branching events, and disregards any information held in the topology. Summary statistics based on branching times are:

56. Crown age, the maximum branching time.
57. Tree height, the maximum branching time plus the root branch length.
58. Gamma, the relative position of internal nodes, with respect to the root and the tips (Pybus & Harvey, 2000).
59. Pigot's rho, the slope of diversification in the first and second half of the tree, corrected for tree size (Pigot et al., 2010).
60. nLTT, the difference in sequence of branching times between two trees; this measures the alignment of diversification between two trees (Janzen et al., 2015).

For independent comparison, the treestats package introduces the function 'nLTT_base', which compares a focal tree with a non-diversifying tree with only two lineages.

2.1.7. Branch lengths

Related to branching times, a number of summary statistics derive from the distribution of branch lengths of a tree, ignoring the associated topology:

61. Phylogenetic diversity, the sum of branch lengths (Faith, 1992; Mendoza et al., 2021).
62. Mean branch length, the sum of branch lengths divided by the number of branches (Janzen & Etienne, 2017).
63. Variance (σ^2) of branch lengths (Saulnier et al., 2017).

Saulnier et al further subdivide branch lengths into two groups: external branch lengths, e.g. lengths of branches attached to the tips of the tree, and internal branch lengths, e.g. lengths of branches only attached to internal nodes of the tree. For these two groups, they then introduce the average and variation as statistics, to end up with:

64. Mean branch length of external branches (Saulnier et al., 2017).
65. Mean branch length of internal branches (Saulnier et al., 2017).
66. Variance of branch length of external branches (Saulnier et al., 2017).
67. Variance of branch length of internal branches (Saulnier et al., 2017).

The ratio between the average length of internal and external branch lengths then provides a measure of the 'stemminess' or 'treeness' of a statistic:

68. Treeness statistic, the ratio between the average internal and average external branch length (Astolfi & Zonta-Sgaramella, 1984).

As a last statistic, we have:

69. Number of tips.

We further introduce a new statistic, explained in the next section:

70. Imbalance steps.

Table 1
Overview of all tree statistics in the treestats package. Included is the classification given by Fischer et al (2023), available size normalization options, and whether the statistics makes assumptions about the tree being ultrametric or binary. The number in the first column refers to the number as used in the main text.

Number	Statistic	Information	Fischer	Normalization	Assumes ultrametric tree	Requires binary tree	Reference
Node statistics							
1	colless	Topology	Imbalance	Yule	NO	YES	Colless, 1982
2	colless_corr	Topology	Imbalance	None	NO	YES	Heard, 1992
3	colless_quad	Topology	Imbalance	None	NO	YES	Bartoszek et al., 2021
4	blum	Topology	Imbalance	None	NO	YES	Blum & François, 2006
5	rogers	Topology	Imbalance	Tips	NO	YES	Rogers, 1996
6	ew_colless	Topology	Imbalance	None	NO	YES	Mooers and Heard (1997)
7	pitchforks	Topology	Shape	Tips	NO	NO	Kendall et al., 2018
8	cherries	Topology	Shape	Yule	NO	YES	McKenzie et al., 1999
9	double_cherries	Topology	Shape	None	NO	YES	Chindelevitch et al., 2021
10	four_prong	Topology	Shape	None	NO	YES	Chindelevitch et al., 2021
11	stairs	Topology	Imbalance	None	NO	YES	Norström et al., 2012
12	stairs2	Topology	Balance	None	NO	YES	Norström et al., 2012
13	il_number	Topology	Shape	Tips	NO	NO	Kendall et al., 2018
14	rquartet	Topology	Balance	Yule	NO	NO	Coronado et al., 2019
15	i_stat	Topology	Imbalance	None	NO	YES	Fusco & Cronk, 1995
16	j_one	Topology	No index	None	NO	YES	Lemant et al., 2022
17	beta	Topology	No index	None	NO	YES	Aldous, 1996
18	avg_ladder	Topology	Shape	None	NO	YES	Kendall et al., 2018
19	max_ladder	Topology	Shape	None	NO	YES	Kendall et al., 2018
20	root_imbalance	Topology	Shape	None	NO	YES	Guyer & Slowinski, 1993
Depth statistics							
21	max_depth	Topology	Imbalance	Tips	NO	NO	Colijn & Gardy, 2014
22	sackin	Topology	Imbalance	Yule	NO	YES	Sackin, 1972
23	average_leaf_depth	Topology	Imbalance	Yule	NO	YES	Shao & Sokal, 1990
24	tot_path	Topology	Imbalance	None	NO	YES	Colijn & Gardy, 2014
25	tot_internal_path	Topology	Imbalance	None	NO	NO	Knuth, 1997
26	var_depth	Topology	Imbalance	Yule	NO	NO	Coronado et al., 2020
27	avg_vert_depth	Topology	Imbalance	None	NO	NO	Colijn & Gardy, 2014
28	b1	Topology	Balance	Tips	NO	NO	Shao & Sokal, 1990
29	b2	Topology	Balance	Yule	NO	NO	Shao & Sokal, 1990
30	symmetry_nodes	Topology	Imbalance	Tips	NO	YES	Kersting & Fischer, 2021
31	max_width	Topology	Balance	Tips	NO	NO	Colijn & Gardy, 2014
32	max_del_width	Topology	Shape	Tips	NO	NO	Colijn & Gardy, 2014
33	mw_over_md	Topology	Balance	None	NO	NO	Colijn & Gardy, 2014
Distance matrix							
34	tot_coph	Topology	Imbalance	Yule	NO	YES	Mir et al., 2013
35	area_per_pair	Topology	Shape	Yule	NO	YES	Lima et al., 2020
36	mpd	Topology + branch lengths	No index	Tips	NO	NO	Webb et al., 2002
37	vpd	Topology + branch lengths	No index	None	NO	NO	Webb et al., 2002
38	j_stat	Topology + branch lengths	No index	None	NO	NO	Izsák & Papp, 2000
39	mntd	Topology + branch lengths	No index	None	NO	NO	Webb et al., 2002
40	psv	Topology + branch lengths	No index	Tips	NO	NO	Helmus et al., 2007
Eigen decomposition							
41	laplace_spectrum_e	Topology + branch lengths	No index	None	YES	NO	Lewitus & Morlon, 2016
42	laplace_spectrum_g	Topology + branch lengths	No index	None	YES	NO	Lewitus & Morlon, 2016
43	laplace_spectrum_a	Topology + branch lengths	No index	None	YES	NO	Lewitus & Morlon, 2016
44	laplace_spectrum_p	Topology + branch lengths	No index	None	YES	NO	Lewitus & Morlon, 2016
45	min_laplace	Topology + branch lengths	No index	None	NO	YES	Chindelevitch et al., 2021
46	max_laplace	Topology + branch lengths	No index	None	NO	YES	Chindelevitch et al., 2021
47	min_adj	Topology + branch lengths	No index	None	NO	YES	Chindelevitch et al., 2021
48	max_adj	Topology + branch lengths	No index	None	NO	YES	Chindelevitch et al., 2021
Network science							
49	wiener	Topology + branch lengths	Shape	None	NO	YES	Chindelevitch et al., 2021
50	diameter	Topology	Shape	None	NO	YES	Chindelevitch et al., 2021
51	max_betweenness	Topology	Shape	Tips	NO	YES	Chindelevitch et al., 2021

(continued on next page)

Table 1 (*continued*)

Number	Statistic	Information	Fischer	Normalization	Assumes ultrametric tree	Requires binary tree	Reference
52	max_closeness	Topology	Shape	Tips	NO	YES	Chindelevitch et al., 2021
53	max_closenessW	Topology + branch lengths	Shape	None	NO	YES	Chindelevitch et al., 2021
54	eigen_centrality	Topology	No index	None	NO	NO	Chindelevitch et al., 2021
55	eigen_centralityW	Topology + branch lengths	No index	None	NO	NO	Chindelevitch et al., 2021
Branching times							
56	crown_age	Branching times	No index	None	NO	NO	
57	tree_height	Branching times	No index	None	NO	NO	
58	gamma	Branching times	No index	None	YES	NO	Pybus & Harvey, 2000
59	pigot_rho	Branching times	No index	None	YES	NO	Pigot et al., 2010
60	nltt_base	Branching times	No index	None	YES	NO	Janzen et al., 2015
Branch lengths							
61	phylogenetic_div	Topology + branch lengths	No index	None	NO	NO	Faith, 1992
62	mean_branch_length	Topology + branch lengths	No index	None	NO	NO	Janzen & Etienne, 2017
63	var_branch_length	Topology + branch lengths	No index	None	NO	NO	Saulnier et al., 2017
64	mean_branch_length_ext	Topology + branch lengths	No index	None	NO	NO	Saulnier et al., 2017
65	mean_branch_length_int	Topology + branch lengths	No index	None	NO	NO	Saulnier et al., 2017
66	var_branch_length_ext	Topology + branch lengths	No index	None	NO	NO	Saulnier et al., 2017
67	var_branch_length_int	Topology + branch lengths	No index	None	NO	NO	Saulnier et al., 2017
68	treeness	Topology + branch lengths	No index	None	NO	NO	Astolfi & Zonta-Sgaramella, 1984
No grouping							
69	number_of_lineages	Topology + branch lengths	No index	None	NO	NO	
Introduced here:							
70	imbalance_steps	Topology	No index	Tips	NO	NO	Janzen & Etienne, 2024

2.2. A new statistic: Imbalance steps

For every reconstructed, binary, phylogenetic tree, we can represent it such that the tree, starting at the crown of the tree, consists of two main branches (crown lineages A and B), from which daughter branches branch off, which in turn might generate their own daughter branches. This looks very much like a classic cladogram.

At one extreme we have the fully balanced tree: assuming that a tree has $2^n = N$ lineages, in each half of the tree we then find N/2 tips, of which $\log_2(N / 2)$ are direct daughters of the crown lineages, and $N/2 - \log_2(N/2)$ are daughter lineages of daughter lineages etc. This yields a fully balanced topology, with the two crown lineages functioning as main 'stems' of the tree (Fig. 1A). At the other extreme we have the fully imbalanced tree, where again we have the two main branches, but now find that all daughter branches are attached to one of the crown lineages, and that all daughter branches are terminal branches, i.e., they do not have daughter lineages themselves (Fig. 1B).

If we keep track of how branches are attached to each other, we can transform the fully balanced tree into the fully imbalanced tree in a stepwise fashion, by re-attaching branches to one of the two crown lineages, until all branches are terminal branches and attached to the same crown lineage.

Then, we define the imbalance steps statistic (statistic 70) as the number of steps it takes to transform a tree into a fully imbalanced tree, where high values indicate a very balanced tree, and low values indicate a very imbalanced tree.

For a fully balanced tree of N tips, the number of steps required to transform the tree into a fully imbalanced tree is $N - (\log_2(N / 2) + 2)$, where $\log_2(N / 2)$ is the number of daughter lineages already connected to the crown lineage accepting re-attached branches, and 2 is the two crown lineages themselves. This simplifies to: $N - \log_2(N) - 1$, and we can use this expression to normalize the observed number of required steps by tree size.

Although the total number of steps required is straightforward to calculate, observing and studying trees of intermediate balance, i.e., that are not fully (im)balanced is less straightforward: there are many different move orders possible to go from a fully balanced to a fully imbalanced tree. Here we explore a number of move algorithms, that can be categorized firstly into two groups, based on the set of potential branches that the algorithm chooses from: (1) group A that re-attaches any branch, and (2) group T that re-attaches only terminal branches (e.g., branches that have no daughter branches). Then, within each group, there are three ways to determine which branch from all available branches to pick: (R) a branch is selected randomly, (Y) the youngest (shortest) branch is selected or (O) the oldest (longest) branch is selected. Combining these two we obtain six selection algorithms (A-R, A-Y, A-O, T-R, T-Y & T-O). However, both TY and AY pick from the same branch set (when selecting the youngest branch, this is always a terminal branch), and thus we obtain five unique selection algorithms: A-R, A-O, T-R, T-Y and T-O. Please note that these algorithms determine the order in which branches are moved, but do not influence the total number of steps required (e.g. the imbalance steps statistic).

2.3. Numerically testing for (im)balance

Strictly mathematically, a balance statistic is defined as a statistic that increases with balance, such that the most extreme value is gained for the most balanced tree (Fischer et al., 2023). Similarly, an imbalance statistic decreases with increasing balance. For many statistics, these properties have been deduced mathematically, however, for some statistics this is not trivial, or impossible. Here, we explore balance properties numerically. By starting from a fully balanced tree, and going in a step-wise fashion towards a fully imbalanced tree, we can numerically verify that the trees at the extreme ends correspond to maximum (or minimum) statistic values. Furthermore, in order to qualify as a proper balance statistic, we require the transition between the extremes to be monotonic.

2.4. Empirical data

To assess correlations across statistics, we assess all summary statistics on a dataset of 221 phylogenetic trees collected on a family level, inspired by similar data used in Condamine et al (Condamine et al., 2019). We started with large trees on the level of orders or clades, and split these up into subtrees on the family level. We collected supertrees for Polypodiopsida (Ferns) (Testo & Sundue, 2016), Angiospermae (Flowering plants) (Zanne et al., 2014), Chondrichthyes (Cartilaginous Fish) (Stein et al., 2018), Actinopterygii (Ray-finned fish) (Chang et al., 2019), mammals (Álvarez-Carretero et al., 2022), amphibia (Jetz & Pyron, 2018)) and birds (McTavish et al., 2024). Using the R package 'taxize' (Chamberlain and Szöcs, 2013), we recovered the associated family per tip in each tree. Then, we split up the super trees based on these family classifications. Per 'family-tree', we then retrieved the described number of species as reported on NCBI (again using the 'taxize' R package), taking care to correct for reported hybrids and local variants. In order to select trees that are closest to 'true' species trees, we only selected trees where the number of tips covered at least 80 % of the described species on NCBI. Lastly, we only used trees where the number of tips was at least 10. This resulted in 122 trees for birds, 45 trees for amphibia, 42 trees for ray-finned fish, 24 trees for mammals, 5 trees for cartilaginous fish, 7 trees for ferns and 7 trees for flowering plants. To correct for relatedness between phylogenies, we created a backbone phylogeny (with families at the tips), to be used for a phylogenetic correction in the statistical calculations we performed. To create such a tree, we first created a distance matrix between all seven taxonomic groups (using age estimates from (Chen et al., 2012; Donoghue & Benton, 2007; Kumar & Hedges, 1998; Testo & Sundue, 2016; Wang et al., 1999)), and converted this distance matrix using an UPGMA algorithm into a backbone phylogeny. Then, at each tip of this backbone phylogeny, we replaced the respective order with the family-level-pruned super tree.

The backbone tree was used to calculate correlations, correcting for tree size and phylogenetic correlation. To obtain PCA estimates we performed phylogenetic corrected PCA using the R package 'phytools' (Revell, 2012). In Fig. 3, the animal silhouettes were added using the R package 'rphylopic' (Gearty & Jones, 2023).

2.5. Simulated data

We explored several aspects of phylogenetic summary statistics by applying them to simulated trees. We simulated trees using a number of different diversification models. Unless stated otherwise, all trees were simulated conditional on having 300 extant tips and a crown age of 10 My. The diversification rate was set to 0.5, with either no extinction and speciation at 0.5, or with extinction set to 0.1 and speciation set to 0.5 + 0.1 = 0.6, thus keeping the rate of diversification the same as without extinction. Extinct species were removed from the tree to yield a reconstructed tree, as in the empirical data.

We explored the following models:

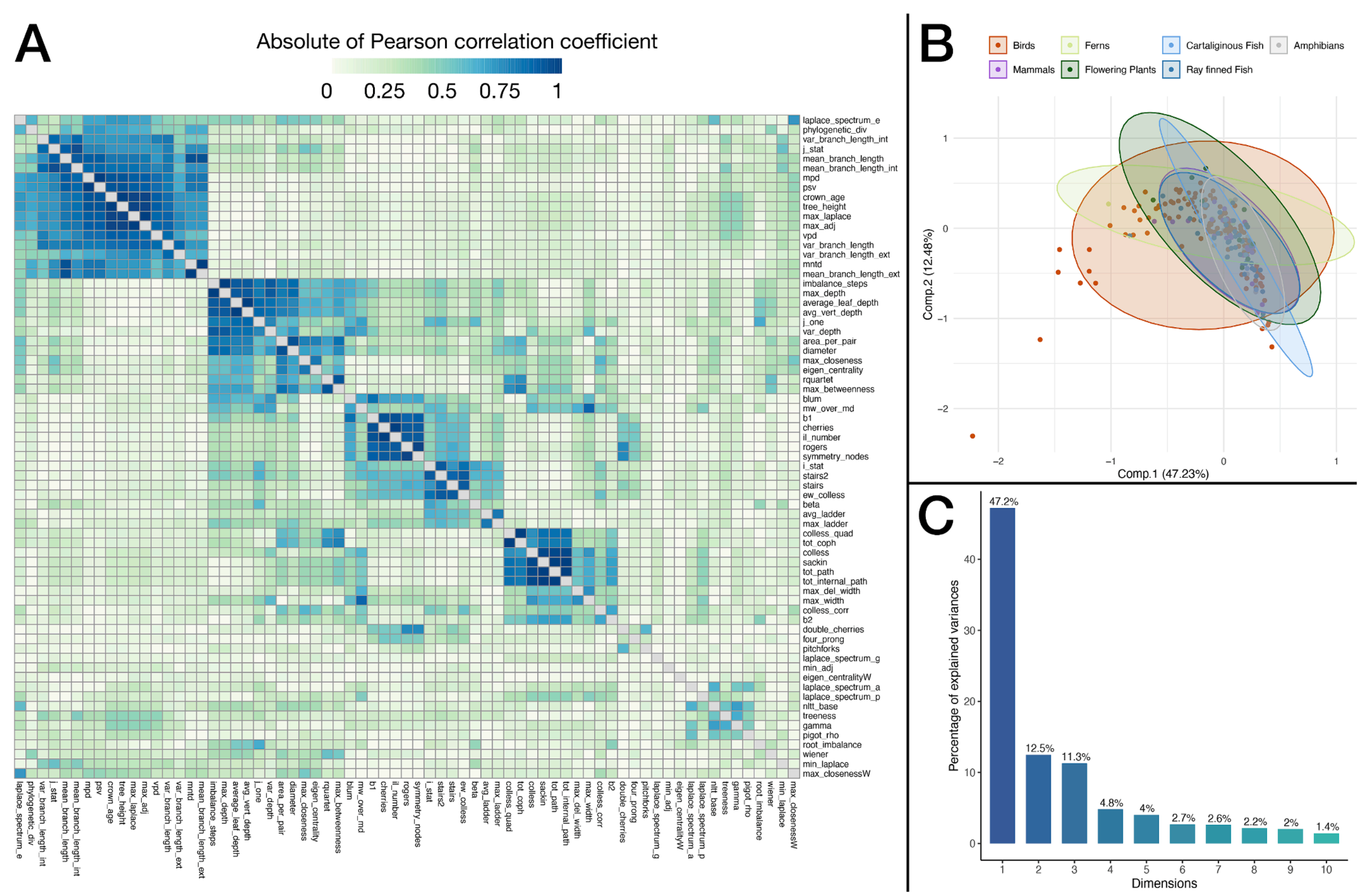

Fig. 2. **A**: Absolute of Pearson Rank Correlation between summary statistics calculated for 221 phylogenetic trees sampled across the tree of life. Correlations are corrected for phylogenetic relatedness and tree size. **B:** First two components of this PCA analysis, where dots are colored according to the respective taxonomic group studied, showing that across the tree of life, phylogenetic tree properties, although variable, tend to share many characteristics. **C:** percentage of explained variance of a PCA analysis of all 70 statistics on the empirical phylogenies, showing that the first five dimensions explain at least 80 % of the observed variance.

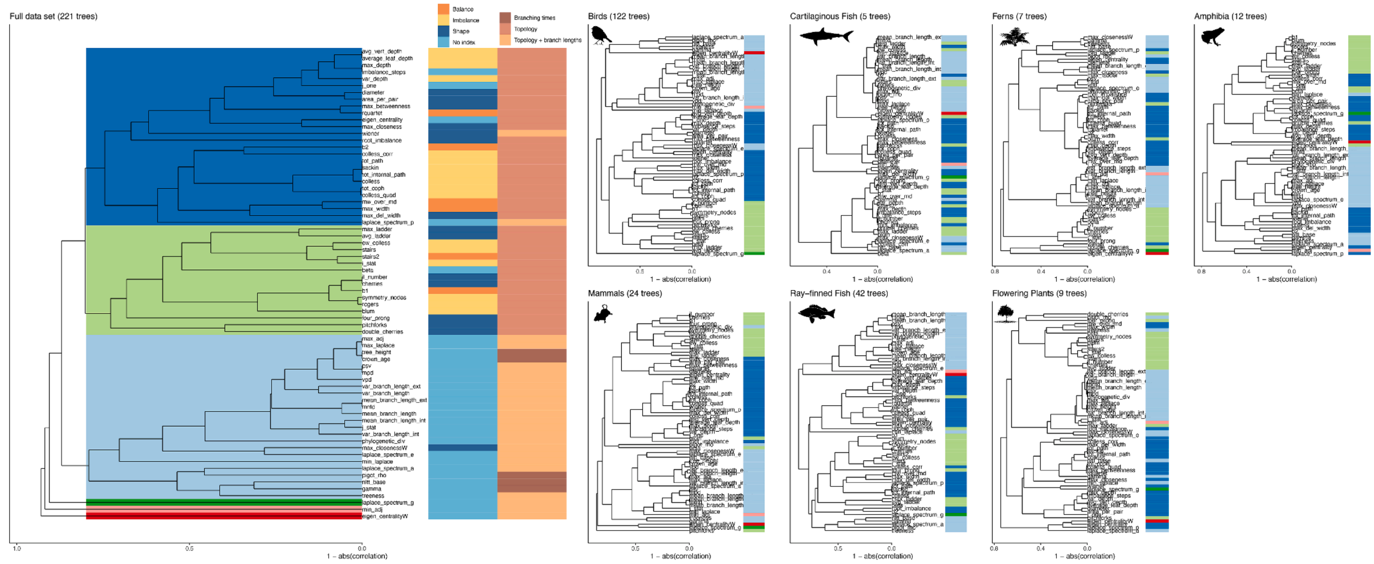

Fig. 3. Correlation dendrograms shown for the entire empirical dataset (left) and for each taxonomic group separately. Correlation dendrograms are based on 1 – abs (correlation). All correlations are corrected for size and phylogenetic signal. Colors spanning the dendrogram show clusters based on a cut-off of a minimum absolute correlation of 0.2. These spanning colors return next to the tips in each of the taxonomic group-based correlation dendrograms, to indicate how well the overall clustering is preserved within each taxonomic group. Colors next to the tips in the entire dataset dendrogram indicate whether the statistic in question is defined as a balance statistic (orange), imbalance statistic (yellow), shape statistic (dark blue) or not topology related (light blue), following (Fischer et al., 2023). The second column indicates the type of information required to calculate the relative statistic (only topology, only branching times, or topology and branch lengths). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

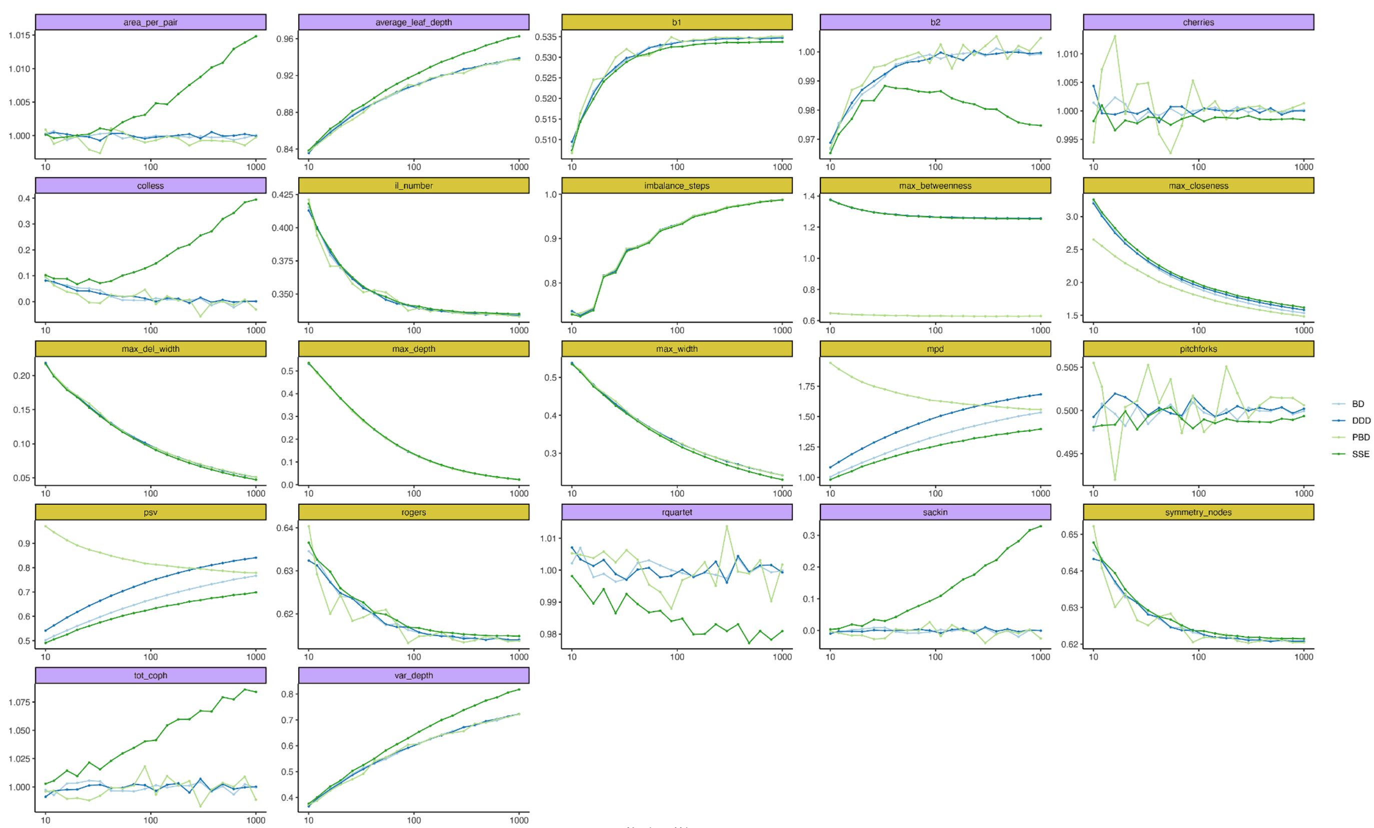

Fig. 4. Observed variation of tree statistics across differently sized trees, for summary statistics that provide corrections for tree size. Gold colored statistics use the number of tips to normalize the summary statistic, purple-colored statistics use the expectation under the Yule model to normalize the summary statistic. Shown are the average tree statistic values across 10.000 replicates per model, per tree size, for four diversification models. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

- Constant Rates Birth-Death (BD) model
- Diversity-Dependent Diversification model (DDD) (Etienne et al., 2012), with K set to 300 + 20 % = 360.
- Protracted birth–death model (PBD) (Etienne & Rosindell, 2012) with λ (speciation completion rate) at 0.1, the incipient speciation rate of good and incipient species at 0.7 and the extinction rate of good and incipient species at 0.1.
- Binary State dependent diversification model (SSE) (FitzJohn, 2012) with b_0 and μ_0 equal to the abovementioned speciation and extinction rate, and b_1 and c_1 equal to 1.5 times b_0 and μ_0. We set the transition rates between states, q_{01} and q_{10}, both equal to 0.1.

Parameter values for each diversification model were chosen in order to emphasize (but not exaggerate to the extreme) the different peculiarities of each model.

2.6. Tree size

To study the effect of tree size and explore the effectiveness of tree size normalization across statistics, we simulated trees with a varying number of tips across the four models, i.e. the number of tips N varied in 10^[1, 1.1, 1.2, …, 3]. We kept the speciation rate constant and varied the crown age in order to allow for the diversification process to be able to reach the target number of tips, where the crown age T was given by: $\frac{1}{\lambda-\mu}\log(N/2)$.

2.7. Correlations

For each of the models, with or without extinction, we simulated 300.000 trees and calculated the corresponding summary statistics. Using these statistics, we calculated the Pearson rank correlation (R) of the summary statistics. We then constructed a distance matrix across summary statistics based on 1 − |R|. Using this distance matrix, we clustered summary statistics together using an unweighted pair group method with arithmetic mean (UPGMA) algorithm.

We performed this analysis for each diversification model separately, but also for the entire dataset, e.g. for all 1.200.000 trees combined (focusing only on trees with extinction rate equal to 0.1).

3. Results

3.1. Empirical data

We find that across the empirical datasets, PCA analysis does not separate the different taxonomic groups (Fig. 2B) based on summary statistics of the used phylogenetic trees, supporting further analysis for all trees together. Furthermore, we find that the first five axes of the PCA analysis explain 79.8 % of the observed variation (Fig. 2C), suggesting careful selection of about five summary statistics might capture the large majority of observed variation across empirical trees. When we break down the contribution of each summary statistic for the first five PCA axes (Figure S1), we find that the majority of variation is explained by the first PCA axis (47.2 % of all variation explained), which is driven by depth and shape associated statistics, such as symmetry nodes, IL number, cherries and pitchforks. The second and third PCA axes (12.5 % and 11.3 % variation explained respectively), are driven by balance-associated summary statistics, such as the Colless statistic, the Colless Quad statistic and the Sackin index. The fourth (4.8 % variation explained) PCA axis then focuses on branching time associated summary statistics, such as gamma, nLTT and Pigot's rho. The fifth PCA axis (4.0

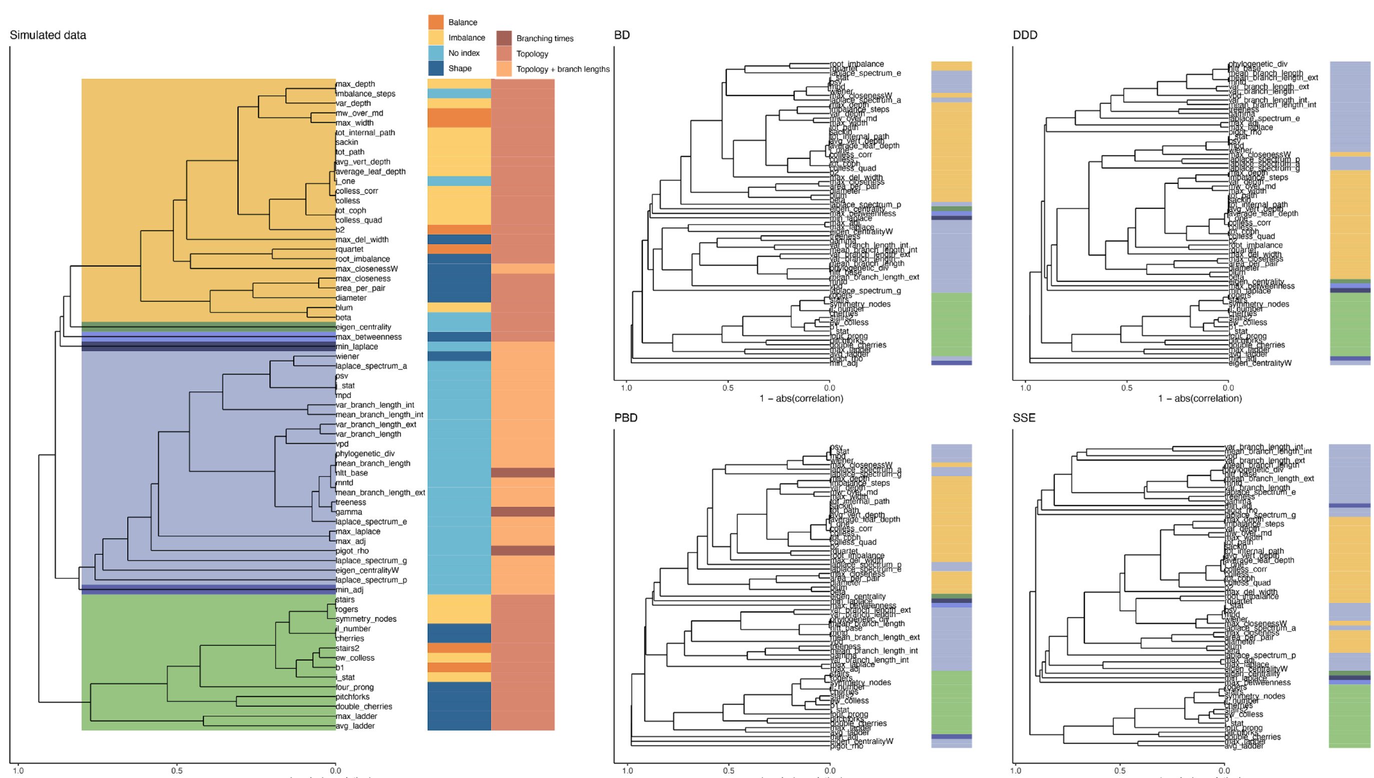

Fig. 5. Summary statistics grouped together based on their correlations across 1,200,000 trees, consisting of 300,000 reconstructed trees simulated per model (BD, DDD, PBD and SSE) with extinction. Colored rectangles indicate statistics grouped together based on a minimum absolute correlation of 0.2; please note that colors are arbitrary and do not correspond to colors in Fig. 3. Colors next to the tips in the entire dataset dendrogram indicate whether the statistic in question is defined as a balance statistic (orange), imbalance statistic (yellow), shape statistic (dark blue) or not topology related (light blue), following Fischer et al. (2023). The second column indicates the type of information required to calculate the relative statistic (only topology, only branching times, or topology and branch lengths). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

% variation explained) is driven by topology associated statistics again, including equal weights Colless, average ladder and the stairs metric.

Using the obtained correlations, we constructed a correlation dendrogram, which groups statistics that share high similarity (i.e. that are strongly correlated, either positively or negatively) together (Fig. 3). We then grouped together statistics into larger clusters when statistics within a group at least had an absolute correlation of 0.2. For all taxonomic groups together, we find three large clusters, grouping together (roughly), all balance indices, all shape indices and all branch length associated indices. The largest cluster (dark blue, Fig. 3) combines balance statistics such as the Colless, J^1 and depth related statistics. Related to this cluster, we recover another topology associated cluster (green, Fig. 3), which mainly includes shape associated metrics such as the cherries index, pitchforks, ladders and stairs associated statistics, but also the beta and Rogers statistic. The third large cluster we recover includes all statistics that take into account branch lengths or branching time, such as mean branch length, gamma and mean pair distance. We recover three statistics that do not group convincingly within these three large clusters: Gap of the distance Laplacian spectrum, minimum eigen value of the adjacency matrix and weighted Eigen centrality. Although across all trees we recover a relatively clear subdivision into clusters based on information used, these relationships break down when we examine them per taxonomic group. Colored bars next to the correlation dendrograms per taxonomic group indicate the clustering found across all trees (Fig. 3), and we find that in most taxonomic groups, the previously recovered clusters are dispersed across the dendrogram, although it seems that the richer datasets (birds, ray-finned fish) experience this less.

3.2. Simulated data: Tree size

Empirical trees are an invaluable source of information, but are all different in size. This tree size variation may cause spurious correlations (Fig. 4). For some statistics, normalization terms are available that correct for this, but we suspected that these may be insufficient. To test this, we simulated trees of different sizes under a range of different diversification models. We find (Fig. 4) that only the cherries, pitchforks and rquartet statistics appear to be able to correct for tree size, regardless of the underlying model. The area per pair, Colless, Sackin and total cophenetic index can also correct for tree size, but only if the model used to simulate the tree matches that for the expectation, which is a Yule model. In all other cases, we find that tree size, even after correction, still correlates with the summary statistic.

3.3. Simulated data: Correlations

To avoid effects of tree size, we simulated trees under different diversification models, keeping tree size (and crown age) constant. Then, we constructed correlation matrices for the different models (Fig. 5). Similarly to the empirical data, we recover three large clusters of statistics. Again, we find that all branching time related statistics cluster together. Similar to in the empirical data, we find two large clusters of topology related statistics, with one cluster including many shape-related statistics (ladder, stairs, cherries), and one cluster associated with depth related statistics and statistics related to the Colless statistic. We now recover four singleton statistics, being the eigen centrality, max betweenness, minimum eigen value of the adjacency matrix, and minimum eigen value of the Laplacian matrix. Interestingly, weighted eigen centrality now groups with branch length related statistics, which is different from how it behaved on the empirical data, possibly due to the constraints used to simulate trees.

Across the four diversification models, we find that the topology cluster including cherries and stairs is retained. Similarly, the cluster including the Colless and Sackin indices is also usually largely retained, although often one or several statistics cluster differently, depending on the diversification model. Lastly, the branching times cluster is often subdivided into two subsections: one including statistics correlating with mean_branch_length and one including statistics correlating with

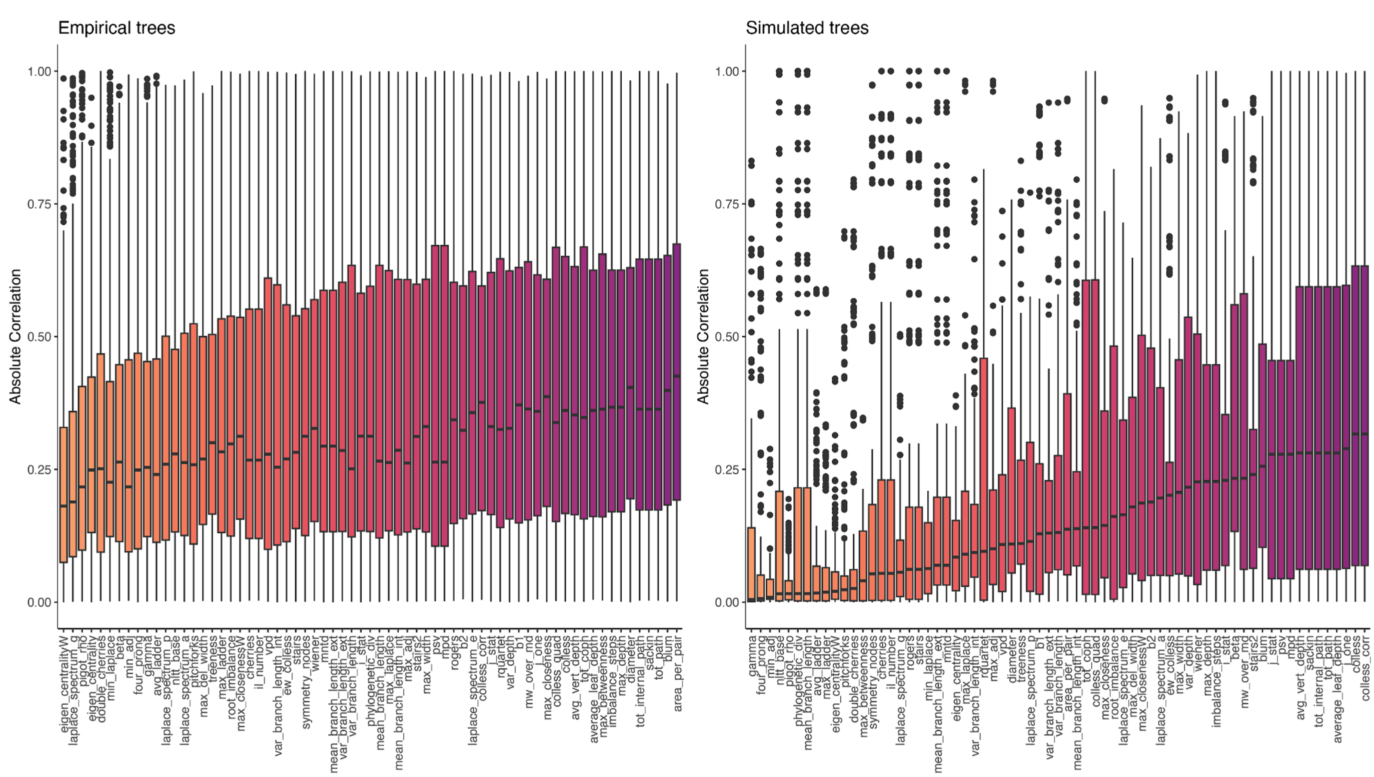

Fig. 6. Absolute correlation coefficients with all other summary statistics, per summary statistic, for empirical trees (left panel, same trees as in Fig. 2) and simulated trees (right panel, same trees as in Fig. 4). Statistics are sorted by their median absolute correlation value. High values indicate that a statistic has high overlap with other statistics, whereas low values indicate that a statistic contains a high amount of unique information about the phylogenetic tree, not captured by other statistics.

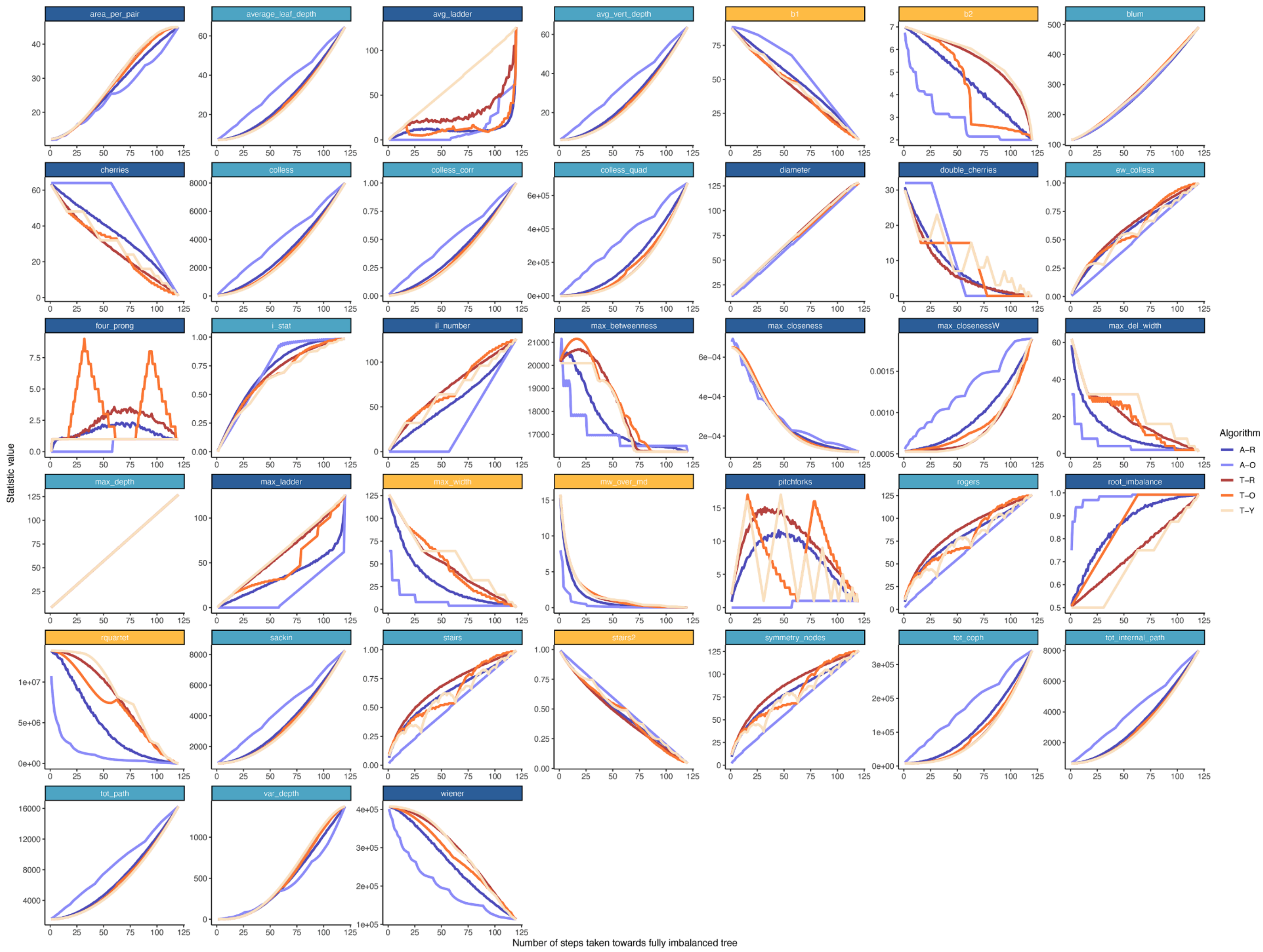

Fig. 7. The average balance related summary statistic value of 100 replicate trees of 128 tips that are increasingly imbalanced using one of six algorithms available. These algorithms move branches around to increase imbalance, using any branch, selecting at random (A-R), using any branch, selecting the youngest branch first (A-Y), using any branch, selecting the oldest branch first (A-O), using only terminal branches, selecting at random (T-R), using only terminal branches, selecting the youngest branch first (T-Y) and using only terminal branches, selecting the oldest branch first (T-O). Colors above the plots indicate whether the statistic in question is defined as a balance statistic (orange), shape statistic (dark blue) or not topology related (light blue), following (Fischer et al., 2023). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the mean pair distance.

3.4. General information content

When we look at how each summary statistic correlates with all others, we find that there is quite a difference between statistics, with some statistics having relatively high average correlation values (Fig. 6), whereas others appear to hardly correlate with any of the other statistics, indicating they contain information not captured by other statistics. For empirical trees, we find that weighted eigen centrality and Pigot's rho perform especially well in this respect, and also three out of four components of the distance Laplacian spectrum associated summary statistics show low median correlation values. For simulated trees, again Pigot's rho show a low average correlation value, outpaced by the gamma, nltt and four prong statistics.

3.5. Intermediate balance

Using the algorithms described to create intermediately imbalanced trees, we explored (im)balance statistics on intermediately balanced trees. We find that almost all statistics vary non-linearly between the two known extreme values (Fig. 7) (e.g. the perfectly balanced and completely imbalanced tree). Only the diameter and max depth statistic show a linear relationship with intermediate balance. Furthermore, we find strong variation across the different imbalancing algorithms, suggesting that capturing intermediate balance is a non-trivial endeavor and that intermediate balance may, depending on the tree topology, lead to very different summary statistic values. We find that first selecting terminal branches as opposed to randomly selecting branches tends to lead to a slower approach of the value of the completely imbalanced tree. The specific type of branch (random, oldest, youngest) seems to matter a lot as well, where selecting older branches causes larger initial deviations. For some statistics (avg_ladder, double cherries, ew_colless, four_prong, max_betweenness, pitchforks, rogers, stairs and symmetry nodes), we find that their relationship with balance is non-monotonic, indicating that statistic values are sometimes associated with multiple degrees of intermediate balance, making these statistics only useful at their extremes.

3.6. Numerically verifying balance

Using the six imbalancing algorithms, we numerically assessed

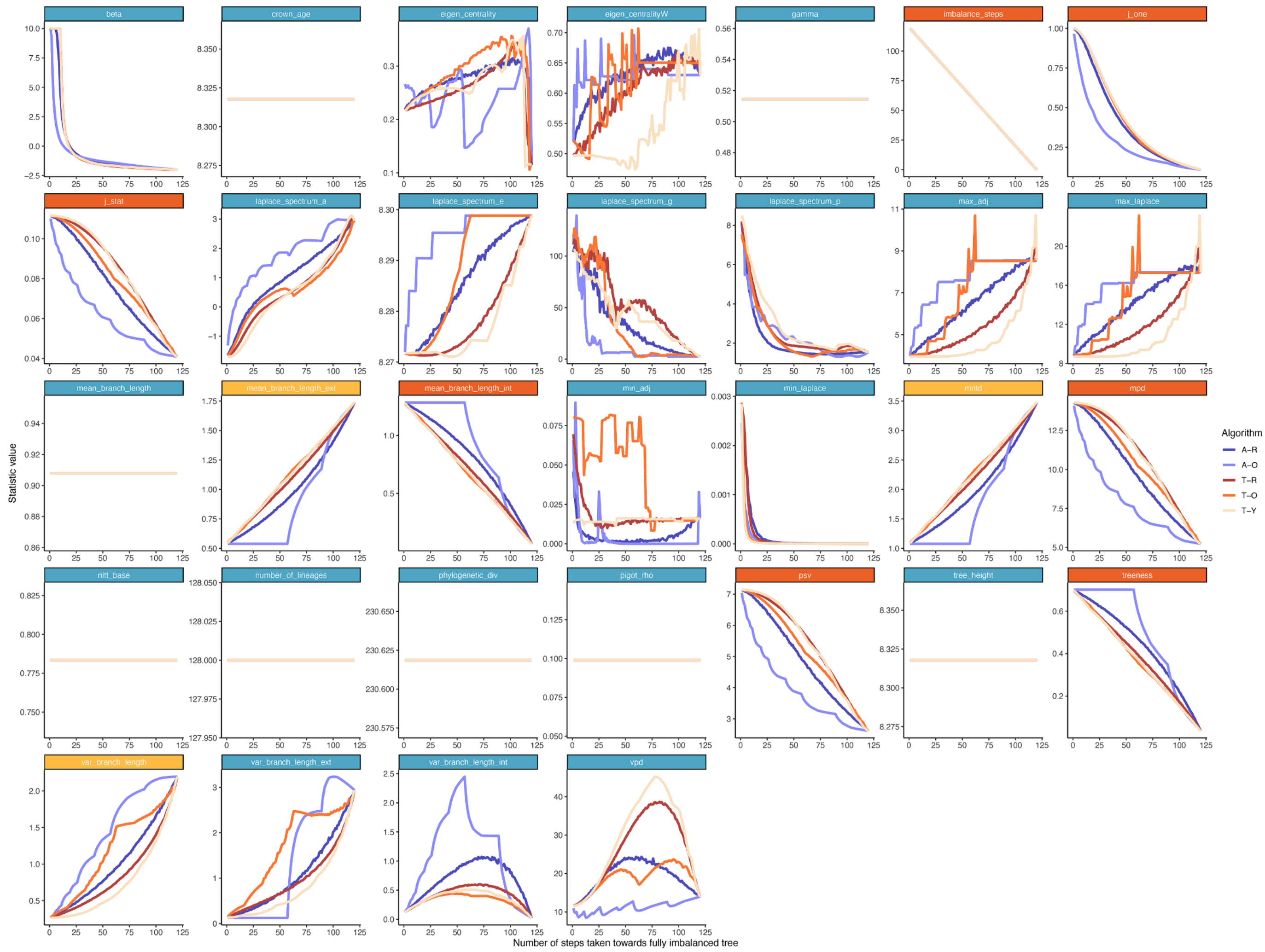

Fig. 8. The average summary statistic value of 100 replicate trees of 128 tips that are increasingly imbalanced using one of six algorithms available (see caption Fig. 7). Colors above the plots indicate whether the statistic in question is numerically assessed as a balance statistic (orange), imbalance statistic (yellow) or has no consistent relationship with balance (light blue). Shown are results for all statistics not considered in Fig. 7. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

imbalance for 32 statistics not included in Fischer et al. (2023). We scored these statistics as (im)balance statistics if they behaved monotonic, and if the extreme values are at the most balance and imbalanced tree. We find (Fig. 8) that the imbalance steps, J^1, J statistic, mean internal branch length, mean pair distance, psv and treeness statistics behave as balance statistics, and that the mean external branch length, mean nearest taxon distance and variance in branch length behave as imbalance statistics. Here we show results only for branching times resulting from a B-D model, but in figure S5 we show that the choice of generating model does not affect these conclusions.

4. Discussion

Here, we have introduced a new R package called 'treestats', that provides fast (Figure S3, S4) and user-friendly code to quickly assess a wide range of phylogenetic summary statistics (Table 1). We have used the package to analyze both empirical and simulated phylogenetic trees and unravel their interdependence. Firstly, we have found that almost all summary statistics are highly sensitive to effects of tree size, even after attempting to correct for tree size. Secondly, we have explored correlations across summary statistics on both empirical and phylogenetic trees, identified several larger clusters of closely related statistics and identified statistics with particularly low information overlap. Lastly, we have explored the variation of summary statistics with intermediately balanced trees, and found several summary statistics that vary non-monotonically with balance.

Many summary statistics correlate with tree size and therefore provide normalization functions to correct for this. However, we find that most of these corrections are inadequate and do not correct sufficiently for tree size. Corrections making use of the maximum value (such as for instance the Imbalance steps, b1 and Roger's J statistic) fail to correct for the impact of the diversification process: for different tree sizes, the diversification process has a different expected summary statistic value, which is often different from the maximum obtainable value. Some other summary statistics recognize this (such as for instance the Sackin, b2 and Colless statistics) and provide normalization options depending on the Yule and PDA process. We find that when the generating process is close to these processes, these normalizations work very well, but that when the generating processes are different, the normalization fails. Across all statistics, we find that only the cherries and pitchforks statistics are able to adequately correct for size, but for these statistics we do not find much variation across diversification models, suggesting low discriminatory power. Therefore, we recommend using tree size as a covariate when comparing summary statistics across trees of different sizes, instead of relying on self-normalization of a summary statistic.

Across empirical phylogenies, we recover three clusters of summary statistics that cover the same information given their high correlation: (1) a cluster centered around balance related statistics such as Colless, Sackin and J^1, (2) a large cluster centered around shape-related statistics such as stairs and cherries and (3) a cluster centered around branch length related statistics, such as mean pair distance and the nLTT statistic. Furthermore, we recover three statistics that do not group within these clusters: distance Laplacian spectrum eigen gap, minimum eigen value of the adjacency matrix and weighted eigen centrality. We do not find that balance or imbalance statistics preferentially correlate with each other, although there does seem to be a divide between statistics that use branch lengths, and those that do not. Some statistics correlate extremely well with each other, often following previous derivations. For instance mpd and psv have been shown to be mathematically identical (Tucker et al., 2017), and total internal path, total path and the Sackin index are also known to be closely related (Fischer et al., 2023). Perhaps a surprising result is that the Sackin and Colless indices are very highly correlated as well, indicating strong information overlap.

Across simulated trees, we recover the same three larger clusters, centering again around balance, shape and branch length related indices. For simulated trees, we recover one extra outlier statistic (in addition to the outlier statistics recovered for the empirical data): max betweenness. Across diversification models we find that these clusters vary considerably, with the exception of the shape cluster that is consistently recovered across diversification models.

Using the average correlation across either taxonomic groups or diversification models, we identify several statistics that have a lower information overlap (on average) than other statistics, making these statistics prime candidates to capture unique information of a phylogenetic tree. For empirical trees, we recover Eigen centrality and Pigot's rho as having low overlap with other statistics, whereas for the simulated trees, we find the gamma and four prong statistics. Interestingly, these are not necessarily the statistics that clustered outside a cluster in our previous clustering analysis. However, in this previous analysis, statistics were grouped together using an UPGMA algorithm, which sequentially groups together statistics based on how close they are. Average values then easily get lost due to the incremental grouping of the clustering algorithm. As such, the average values strongly complements the clustering analysis, but does not provide context as to which other statistics are closely related to the focal statistic. Therefore, we suggest to use both sources of information when choosing a particular statistic. Interestingly, shape-related statistics in particular appear to have low average correlations in simulated trees, which could be due to the uniform nature of the trees simulated, i.e. all trees had the same number of lineages and crown age, possibly masking other statistics.

An important aspect that is measured by many statistics, is the overall (im)balance of a phylogenetic tree. However, such statistics are typically defined and explored only for completely balanced or completely imbalanced trees and their relationship with changes in balance remains understudied. Here, we have introduced several algorithms to generate trees of intermediate balance in a systematic way. We found that most statistics do not vary linearly with balance, and for some statistics even found a non-monotonic relationship with balance, indicating that these statistics cannot be reliably used to compare balance of phylogenetic trees; these include the avg_ladder, double cherries, ew_colless, four_prong, max_betweenness, pitchforks, rogers, stairs and symmetry nodes statistics. The non-linear relationship between balance and summary statistic value could also be due to the way we created trees of intermediate balance. Indeed, we found for some statistics strong variation across different imbalancing algorithms, e.g., for the maximum ladder statistic there is an almost linear relationship with balance if the intermediate balance is generated using terminal tips, starting with the youngest tips first, but a very non-linear relationship if random tips are picked. We draw two conclusions from our analysis of intermediate balance: firstly, almost all statistics have a non-linear or complex relationship with balance, making interpretation of intermediate summary statistic values difficult. Secondly, the strongly differing relationships between summary statistic value and balance suggest that balance is not a one-dimensional property of a phylogenetic tree, but contains multidimensional information. This is in accordance with the previous subdivision into multiple correlated clusters of summary statistics we recovered for both empirical and simulated trees. The analysis of (im)balance of a phylogenetic tree therefore remains a challenge that is not easily defined except at its extremes, and future explorations leading to a better understanding of the relationship between balance statistics and intermediately balanced trees are highly warranted.

Using our imbalancing algorithms, we have also numerically explored the relationship with imbalance for statistics that take into account branch length information. We find that the J statistic, the mean internal and external branch length, mean nearest taxon distance, mean pair distance, treeness and variation in branch length all behave similarly to topology-based (im)balance indices. We hypothesize that this is due to the ultrametric nature of the trees we consider here: given an ultrametric caterpillar tree, the longest branch is attached to the deepest node, the second longest branch to the second-deepest node etc. This maximizes the mean-pair distance, the mean internal branch length and the variation in branch length. Contrastingly, for the fully balanced tree, distances to the nodes connected to the tips are minimized (again, assuming an ultrametric tree), explaining minimal values for mean-pair distance, mean internal branch length and variation in branch length. Furthermore, we confirm the use of the imbalance steps and J^1 statistics as imbalance statistics. The extension of analysis of balance of statistics provided through this numerical test provides an important extra test when using summary statistics: those summary statistics that can be numerically confirmed to behave as (im)balance statistics may be more informative and preferred over those that are not.

Across our analyses, we have found that different Eigen decompositions of the adjacency matrix, Laplacian matrix and distance Laplacian matrix appear to hold unique information often not contained in other statistics. For instance, different properties of the distance Laplacian spectrum often did not closely cluster together, but instead correlated with very different properties of phylogenetic trees, indicating that these measures of the distance Laplacian spectrum reflect different underlying properties: The principal eigenvalue typically clustered with balance statistics, whereas the asymmetry and peakedness clustered with branching time related statistics. The eigengap measure was a strong outlier for empirical trees, but clustered with branching time related statistics for simulated trees. Similarly, we also found that eigenvalues and eigenvectors of the adjacency matrix and Laplacian matrix, and also the Perron-Frobenius eigenvector of the adjacency matrix (e.g. Eigen centrality) all had very unique properties and did not overlap strongly with many other statistics. In summary, all these Eigen decomposition associated statistics are promising candidates for comparing and assessing phylogenetic trees. However, in contrast to more 'traditional' statistics, these measures are not necessarily directly connected to specific properties of these trees, making interpretation difficult. Future work connecting these properties to specific tendencies or relationships within trees would empower these statistics to become important tools in a phylogeneticist's toolbox.

Previously, Tucker and colleagues identified three pillars of summary statistics (stemming from previous work in community assembly theory (Pavoine & Bonsall, 2011)): statistics measuring richness, divergence and regularity. They identified three 'anchor' statistics for these pillars: phylogenetic diversity, mean pair distance and variance in pair distance respectively. These three statistics rely heavily on including information from branch lengths, and in our results, we recover that these statistics consistently cluster together amongst other branch length related statistics. This suggests that a large part of variation across trees is missed by selecting these 'anchor' statistics, and that inclusion of statistics that include topology related information would be warranted.

In developing the R package 'treestats', we have taken great care to create a package that only contains rigorously tested code (98 % of all code is tested), and that provides functions that are much faster than currently available (Figures S3 and S4 show 'treestats' is often several orders of magnitude faster). Long term support for software packages is known to often be troublesome (Mangul et al., 2019). Here, we aim to improve long term support using a multi-pronged approach: 1) in developing the R package, we have aimed at minimizing dependencies on other packages, making the package only dependent on three packages, all available on CRAN: the ape package (Paradis & Schliep, 2019) to handle phylogenies, the Rcpp package (Eddelbuettel, 2013; Eddelbuettel & Francois, 2011) to intertwine C++ and R code, and the nloptR package (Johnson, 2007) that performs maximum likelihood optimization for the Beta statistic. Reducing reliance on dependencies reduces potential maintenance issues in the future. 2) We provide ongoing support on https://github.com/thijsjanzen/treestats, which allows for user feedback, bug reporting and for users to suggest inclusion of newly developed tree statistics, for instance through a pull request. 3) We have archived the last version of the R package on three locations: it is available on https://github.com/thijsjanzen/treestats, on https://CRAN.R-project.org/package=treestats (https://doi.org/10.32614/CRAN.package.treestats) and on Zenodo (https://doi.org/10.5281/zenodo.12748170).

When using phylogenetic summary statistics to compare phylogenetic tree properties across different scenarios, models or taxonomic groups, selection of focal summary statistics is crucial to ensure a proper comparison. With our analysis here we have aimed to provide some guidance into which statistics to select. To reduce intercorrelation between statistics, we recommend to pick statistics from the different larger clusters, for instance by selecting the Colless statistic, number of cherries and phylogenetic diversity. Furthermore, we recommend to pick at least one of the statistics that has been shown to show low correlation with all other statistics, such as weighted eigen centrality and Pigot's rho. When aiming to capture the balance of a tree, given our results on intermediately balanced trees, we recommend using the max_depth statistic, as this varies linearly between the most balanced and most imbalanced tree, in contrast to many other balance statistics that have a non-linear relationship with intermediate balance. More generally, when choosing statistics, picking multiple statistics out of those sets of statistics that are known to closely correlate with each other should be avoided. With these limitations in mind, it may be complicated to select the right combination of statistics, also taking into account that often intercorrelations of statistics may depend on the tree generating model, which may not always be known. Alternatively, we are confident that the computational speed of the treestats package can be leveraged by the researcher in order to explore a larger number of statistics and make an informed choice about summary statistic selection.

CRediT authorship contribution statement

Thijs Janzen: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Rampal S. Etienne:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data statement at end of manuscript, see attached files.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ympev.2024.108168.

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